AKANKSSHA ADEPU

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PROFILE

Computer Science Engineering graduate specializing in applied machine learning and AI, with hands-on experience across classification, regression, deep learning (ANNs), and NLP-based recommender systems. Comfortable owning the full pipeline — data extraction, EDA, feature engineering, model building, evaluation, and deployment via FastAPI and Streamlit. Strong foundation in Python, statistics, and model evaluation, with growing interest in GenAI and LLM-based application development.

KEY PROJECTS

Sepsis Early Prediction using Artificial Neural Networks (ANN) | *Python, TensorFlow, Keras, FastAPI*

- Built a deep learning classification model using **TensorFlow/Keras** to predict early onset of sepsis from a Kaggle clinical dataset, framing it as a high-stakes, recall-sensitive prediction problem.
- Performed data cleaning, feature engineering, and class-imbalance handling to optimize **recall** on the positive (at-risk) class — prioritizing minimizing missed cases over raw accuracy, as appropriate for a clinical early-warning use case.
- Deployed the trained model via a **FastAPI** backend, exposing a REST endpoint for real-time prediction requests.

Swiggy Delivery Time Prediction | *Python, Scikit-learn, XGBoost, Streamlit*

- Built and benchmarked 5 ML algorithms (Linear Regression, Random Forest, Gradient Boosting, Decision Tree Regressor, XGBoost) to predict delivery time; selected XGBoost as the best performer after feature engineering and hyperparameter tuning.
- Deployed the final model as an interactive **Streamlit** app, with the full pipeline versioned on GitHub.

Credit Card Fraud Detection System | *Python, Scikit-learn*

- Built ML classification models (Logistic Regression & Random Forest) on imbalanced financial transaction data, applying iterative feature engineering to improve **recall on the minority (fraud) class**.
- Produced visualized prediction reports and a feedback loop for continuous model evaluation and improvement.

Cryptocurrency Market Analysis & Volatility | *Python, Pandas, NumPy*

- Extracted and cleaned data on thousands of digital assets; ran **exploratory data analysis (EDA)** to surface volatility and risk patterns across the market.

Netflix & Book Recommendation Systems | *Python, TF-IDF, Cosine Similarity, Streamlit, FastAPI, Hugging Face*

- Built content-based recommenders using **TF-IDF vectorization and cosine similarity** over show/movie and book metadata.
- Deployed one via Streamlit and one via a **FastAPI** backend, with models hosted on **Hugging Face** — hands-on exposure to building and serving an API layer around an ML model.

TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, XGBoost, TensorFlow, Keras, Classification, Regression, Deep Learning (ANN, CNN), Recommender Systems (TF-IDF, Cosine Similarity), Model Evaluation

Data & Analytics: Python (Pandas, NumPy), SQL, EDA, Statistics

Deployment & Tools: FastAPI, Streamlit, Hugging Face, Git/GitHub, OpenCV, MySQL

Programming: Python, Java

EDUCATION

B.E. — Computer Science Engineering | Keshav Memorial Engineering College, Hyderabad, 2025 | CGPA: 7.65 / 10

Intermediate (MPC) | Vishra Junior College, Hyderabad | 84%

Secondary Education (CBSE) | Akshara International School, Hyderabad | 83%